

Human Service System Planning: 211 Data and Secondary Prevention Services

Michael Stiehl, M.S,^{a*} and Colin J.G. Cepuran, PhD^a

^a*Chapin Hall, Chicago, IL, U.S.A.*

*Corresponding Author. mstiehl@chapinhall.org C/O Impact House, 200 W. Madison Street, Chicago, IL 60606

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The Family First Prevention Services Act (“FFPSA”) was meant to shift the child welfare system toward prevention services. Implementing that shift requires data capable of facilitating secondary prevention system planning. In this manuscript, we argue that such data should be timely, comprehensive, and specific enough to facilitate the identification of emergent needs in any part of a community. We introduce data from 211 Maine that may be fit for that purpose. 211 Maine collects daily data on community requests for assistance. These data are tabulated at the zip code level and are classified in a detailed taxonomy of need types. In a series of descriptive tests, we demonstrate that these data are reasonably comprehensive of Maine’s geography and population and that they reflect expected seasonal and longitudinal patterns. In inferential analyses, we present additional evidence of validity by demonstrating that variation in zip-code level request volume corresponds predictably to measures of socioeconomic vulnerability. Taken together, these findings present evidence of the applicability of 211 data for human services secondary prevention service planning.

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Introduction

The Family First Prevention Services Act (Family First) is one of the most significant pieces of child welfare legislation passed in recent history. Enacted in February 2018, it represents a fundamental shift in federal child welfare policy that, among other provisions, enables state child welfare systems to receive federal title IV-E reimbursement to provide evidence-based programs (EBPs) and services to prevent children from entering foster care (Bipartisan Budget Act, 2018).

The legislation allows jurisdictions to seek Title IV-E reimbursement for EBPs provided to children, parents, and/or kin caregivers that address individual- and family-level contributors of child welfare involvement in three service areas: mental health treatment, substance abuse prevention and treatment, and in-home parent skill building.

This shift toward prevention requires state and county-based child welfare systems to think differently about their local service array. The law's focus on preventing entry by addressing the contributors of child welfare involvement makes it essential that child welfare systems understand what is driving system entry in their community. Only by doing this can children and families be supported by a local service array that is tailored to meet their needs.

To understand these potential drivers, we draw on foundational research into prevention. Since the mid-1960s (Leavell & Clark, 1965) researchers have conceptualized at least three levels of prevention: primary, secondary, and tertiary. Fuller provides a succinct definition of these levels: "Primary prevention would prevent the emergence of a problem, secondary prevention would refer to working on a problem in its early stages, and tertiary prevention would limit the damaging effects of a problem already established" (1989, p.9). These levels of prevention rely on corresponding data

to effectively develop appropriate prevention strategies (Capacity Building Center for States, 2021).

Currently, child welfare systems connect families to services only after they have been reported to, or investigated by, a child welfare agency. Researchers note that a significant portion of those investigations could be avoided through preventative human service delivery (Puls *et al.*, 2021). However, due to statutory obligations, data collected by child welfare agencies capture maltreatment that has already happened. Therefore, these data are only relevant to tertiary prevention strategies.

Other facets of the human service system resemble child welfare in collecting data on the outcomes of a process it would like to prevent. These types of metrics are known as lag indicators and were first popularized by Kaplan and Norton (Harvard Business School, 1996). Lag indicators are pervasive throughout human service systems. A review of the 2022 National Survey on Drug Use and Health, conducted annually by the Substance Abuse and Mental Health Services Administration, for the U.S. Department of Health and Human Services, perfectly illustrates this point: of the forty-nine indicators included in the report all of them are lag indicators.

Human service systems use lagging indicators for several reasons, including accountability, the ease of measuring past performance, and the availability of retrospective data. These indicators, such as reported cases of maltreatment in child welfare or clinical outcomes in health care, provide clear and quantifiable measures of past events, which are essential for documenting effectiveness and regulatory compliance. This reliance on lagging indicators can, however, impede the development of proactive prevention strategies. By focusing on past incidents, human service systems

may overlook the importance of predictive and leading indicators that could help anticipate future problems.

The prevalence of lagging indicators across various human service systems, regardless of country or specific service focus, is evident from multiple studies. Mathews et al. (2020) highlight the heavy reliance on retrospective data in U.S. child welfare systems to measure past incidents of child maltreatment, illustrating the use of lagging indicators for assessing and improving service outcomes. Similarly, a review by Ramalho et al. (2019) of primary health care quality indicators in the UK, categorizes a significant portion of them as lagging indicators. If human service systems are to successfully transition from treatment to prevention, it is clear from these studies that much of the information they currently rely on will be insufficient.

An additional complication of the transition to prevention in human service systems is evident within the FFPSA legislation itself. FFPSA's emphasis on target populations and EBPs mean that most of the prevention activities under the law would fall into tertiary prevention. We argue that, to build truly preventative strategies, child welfare agencies will increasingly need to collaborate with other human service agencies. To keep children safe within their families and communities, leveraging an array of interventions across human service systems will be essential. (Center for States, 2022).

A recent report from the American Bar Association (ABA) reinforces this point. The report's authors discuss how the initial implementation of FFPSA has encouraged states to utilize kinship navigator programs, and other supportive secondary prevention services across state human service systems, to provide comprehensive care and support for families in need (American Bar Association,

2022). These types of cross-system collaborations will be necessary if child welfare is to be successful in preventing the entry of children and families into their system.

To plan and build the type of secondary prevention service array necessary to support the targeted interventions contained within FFPSA, human service systems will need new sets of data that capture the secondary prevention needs of children and families at the whole population level. We argue that these data should have the following characteristics:

- (1) Data must be population level.
- (2) Data must be timely and frequently updated.
- (3) Data must be geographically comprehensive of the jurisdiction it represents.
- (4) Data must be able to define the needs of the whole population.
- (5) Data must be able to generate actionable policy responses.

Data with the above characteristics can be used to identify emergent population-level pressures that child and family supporting systems can address with effective secondary prevention strategies. Without these data, child and family supporting systems will be forced to continue to rely on lagging indicators that facilitate the mitigation—rather than prevention—of problems.

In this paper, we introduce data that has been underutilized in the development of secondary prevention services in child and family supporting systems: 211 service requests. We received 211 service request data, with the support of Casey Family Programs, as part of a partnership between 211 Maine and the Maine Office of Child and Family Services (OCFS). OCFS obtained these data with the goal of identifying gaps in family-strengthening services and programs. The 211 data are available at the

Zip Code Tabulation Area (hereafter “ZCTA”) level in 43 states, for at least some, if not all, of those states’ counties. For many of these states this data goes back multiple years, allowing for trend analysis. Moreover, because these data are collected anonymously at the ZCTA level (211 Counts, 2024), they can be accessed by researchers and practitioners with few restrictions. In this paper, we present evidence that these data accurately, promptly, and comprehensively capture community needs that state and county human service systems will need to effectively build secondary prevention systems.

What follows will draw on child welfare and human services research to further elaborate on our criteria for secondary prevention data in human services. We will introduce the 211 Maine data and conduct tests to demonstrate its fitness for secondary prevention system planning. These tests will include a basic descriptive analysis and statistical tests to assess the 211 data’s comprehensiveness, timeliness, and capacity to generate actionable policy insights. We will conclude by discussing potential applications of these data in the human services context.

Data

We examine the 211 data alongside data from the American Community Survey (ACS). The ACS is a probability survey of Americans (roughly 1% are interviewed) introduced by the Census Bureau in the mid-2000s to generate richer demographic data than the Decennial Census (Treat *et al.*, 2014). Most importantly for our purposes, the ACS is aggregated to produce estimates of the socioeconomic and demographic characteristics of different U.S. geographies. To maximize comparability with 211 data, we utilize estimates of ZCTA Tabulation Area-level (ZCTA) characteristics (the finest-grained

geography at which 211 requests are aggregated). We collect ACS data from Walker and Herman's tidycensus package (2024).

Methodology

We begin exploring the 211 service request data by examining two of the criteria identified above: the data's population and geographic coverage. We do so by assessing the geographic distribution of total requests at the ZCTA level. Each call to 211 Maine can have multiple requests. Thus, for this analysis we focus on request-level data to represent the overall level of need in each ZCTA. To capture geographic comprehensiveness, we produce two analyses. First we produce a Pareto chart, which is meant to measure the distribution of 211 service requests relative to the distribution of the population in Maine. Second, we calculate a Herfindahl-Hirschman Index, a statistic derived from antitrust economics that examines the concentration of an industry, which has been adapted for other disciplines. Each of these methods enable us to quantify the geographic concentration of 211 requests.

Next, we examine whether the data are timely and frequently updated. No specific statistical tests are needed to demonstrate the data's timeliness: 211 requests are tabulated less than 24 hours after a call is made (211 Counts ME). In addition, we conduct a holistic examination of the service categories at which the 211 call data are tabulated to determine whether the calls are sufficiently fine-grained to facilitate a policy response.

Finally, to demonstrate these data's ability to (a) define the needs of the entire population and (b) generate actionable policy insights, we present the results of a series of inferential analyses. These analyses draw on data on community characteristics from

the ACS (see above) to examine whether variation in 211 request volume corresponds with indicators of community vulnerability. We use those community characteristics as predictor variables in two types of regression models. First, we examine how community characteristics at the ZCTA level predict logged 211 request volume using OLS regression. Second, we use those same variables in a geographically-weighted regression (GWR) model to allow the relationships between predictors and request volume to vary across ZCTAs.

In keeping with the best practices for optimizing a Geographically Weighted Regression (GWR) model (Fotheringham, Brunson, & Charlton, 2002), we set the kernel size to 0.92 after cross-validation (conducted in `spgwr`, see: Bivand et al., [2023]), ensuring a balance between model complexity and predictive accuracy by minimizing prediction errors, effectively capturing spatial heterogeneity, and reflecting the appropriate spatial scale for the processes in our study area (Fotheringham, Brunson, & Charlton, 2002).

Next, we chose an adaptive bandwidth set to 0.5 to vary based on data point density, which maintains the statistical reliability of local estimates and prevents overfitting in densely populated areas and underfitting in sparse ones, enhancing the model's ability to accurately capture spatial variability. Additionally, we selected weights computed with a bisquare functional form to prioritize nearby points while reducing the influence of distant ones, thereby improving the reliability of local estimates and providing a balance between sensitivity and noise reduction.

Results of the GWR were presented to partners in Maine with deeper knowledge of the community context (see below) in order to ensure that the statistical results matched known community conditions and to fine-tune the model where necessary.

(Figure 1 about here)

Analysis

We begin by examining the extent to which 211 calls have acceptably broad geographic coverage in Maine. As can be seen in Figure 1, the results of this analysis show 211 requests are overrepresented in high population ZCTAs. The top 25 largest ZCTAs in Maine by population represent 36% of the state's residents and 51% of its 211 requests. Similarly, we found that the total population within ten miles of Maine's largest cities, Bangor, Lewiston, Portland, and Waterville, encompasses 29% of the state's residents but 38% of all the 211 requests for services. However, requests are present in nearly all ZCTAs in Maine.

(Figure 2 about here)

Similarly, requests have a low Herfindahl-Hirschman Index score of 170.35.¹ The distribution of calls across ZCTAs—boundaries of which are compiled from Walker & Rudis (2024) can be seen in Figure 2. Figure 2 corroborates our findings above, 211 requests are concentrated around urban centers and their neighboring areas. There is also some evidence of calls clustered around the eastern and western borders of the state.

The low Herfindahl-Hirschman Index may be the result of all but two ZCTAs in Maine having at least one request, including some zero-population ZCTAs. Requests from zero-population ZCTAs are likely the result of Maine having a robust tourist industry: some communities lack population when the United States Decennial Census

¹ A Herfindahl-Hirschman Index ranges from 0 (perfectly unconcentrated) to 10,000 (perfect concentration—here, all calls coming from a single ZCTA).

is taken in April but do have population in the summer and fall months of the year. The results of these analyses show that 211 requests are overrepresented in high-population ZCTAs. However, requests are distributed across nearly all ZCTAs in Maine.

(Figure 3 about here)

Next, we examine the extent to which calls are updated in a timely and frequent manner. Figure 3 shows calls by day of the week in each month and year. This figure demonstrates that 211 request data capture variation in demand for services by day of the week and month of the year in addition to annual trends. Calls tend to peak consistently year over year, month over month, mid-week on Tuesday, Wednesday, and Thursday. Calls reach their lowest levels over the weekend and into Monday.

(Figure 4 about here)

Among other taxonomies, 211 calls are grouped into the AIRS taxonomy (211 Taxonomy, N.D.), a prominent national ontology of social work needs used by 211 providers across the U.S. Figure 4 depicts a six-month rolling average of the four most common request categories. Distinct variation is present in the data by category, allowing for meaningful understanding of need by request type. As seen in the graph, COVID-related requests dominate, coinciding with the initial and Omicron waves of COVID-19. Next, seasonal trends in socioeconomic pressure (specifically, increased calls related to utilities during winters) are also present. Finally, longer-term secular trends (namely, steadily mounting housing costs—see: MaineHousing, 2022) are also visible in the data.

Thus, we argue that the 211 data captures both short-term and long-term variation in service needs. Calls vary predictably by day of week and month, with Tuesday-Thursday seeing the highest call volumes. The data also reflects major events like COVID-19 waves as well as seasonal needs and long-term trends in housing costs. In addition to demonstrating the ability of 211 data to identify longitudinal trends in community needs, this finding provides strong evidence that 211 service request categories conform to our expectations.

From there, we examine the extent to which 211 data capture community needs in sufficient granularity to facilitate concrete policy responses. There are over 1,900 taxa into which requests are categorized, ranging from the most common (“COVID-19 control”; 53,593 calls) to 410 taxa which received a single request (including “contraception,” “gastroenterology,” and “pet care supplies.”). The requests include a great deal of variation within more general categories. For example, requests categorized as pertaining to addiction are grouped into 13 taxa, with a total of 509 requests. Specific taxa include support groups, internet support groups, and gambling addiction support groups. Requests related to addiction do *not* include other, related categories like substance use (about 53 taxa), alcohol (10 taxa), and opioids (5 taxa).

(Table 1 about here)

We conclude our examination of the extent to which 211 Maine data are sufficiently fine-grained to facilitate targeted policy responses by examining the relationship between 211 request volume and community need. Table 1 shows the result of the first of the models specified for this purpose. It presents the results of a regression of predictors of social determinants of health on ZCTA level logged total 211 service requests. The significant ($p < .05$) and positive predictors are: the number of people living at their current address for more than a year, the number of people on SSI, the number of housing units built in 1939 or earlier, the number of housing units using wood heating, the number of people with a long commute, the number of people whose housing costs exceed 30% of household income, and the number of housing units valued at less than \$100,000. The significant negative predictors are the number of people on public assistance, and the number of housing units using oil heating. Cross-sectional modeling shows that economic deprivation, rurality, and housing quality are significant predictors of higher 211 request volume, even when controlling for population. This provides evidence that requests reflect underlying community socioeconomic factors.

(Table 2 about here)

Our final model allows the relationship between community characteristics and request volume to vary across geography. We do so by conducting a geographically weighted regression (GWR) with identical data to that analyzed in Table 1. Summary statistics for the coefficient estimates produced in that regression are displayed in Table 2. Local R^2 s, which demonstrate geographic variation in model fit, are depicted in Figure 5.

(Figure 5 about here)

Geographically weighted regression reveals that the relationship between community characteristics and requests varies across different regions of Maine. This points to localized factors shaping demand for services, suggesting that these data exhibit sufficient geographic variation to facilitate secondary prevention.

Discussion

In the above analyses, we sought to examine the extent to which 211 data were: population level, timely and frequently updated, geographically comprehensive, capable of defining the needs of the whole population, and able to generate actionable policy responses.

First, we presented evidence that 211 data captured needs at the population level. While concentrated in population centers, requests originate from nearly all ZCTAs. Moreover, the lack of perfect uniformity of request volume across ZCTAs reflects the unevenness of population distribution across ZCTAs.² A measure of concentration derived from antitrust economics indicate that request volume are not highly concentrated to a few ZCTAs—even without accounting for Maine’s relatively concentrated population. While there is no single threshold above which requests would be unacceptably concentrated, we argue that these requests are sufficiently broadly distributed to capture population-level needs.

Next, we examined the extent to which 211 data were timely and frequently updated. The data is frequently updated and reflects fluctuations in demand on multiple timescales. Concretely, data are updated every twenty-four hours. Moreover, we

² ZIP codes facilitate mail delivery rather than the tabulation of population (c.f. Census Tracts), so some ZIP codes have few to no residents.

presented evidence that request volume reflects weekly (peaking mid-week) and seasonal trends (reflecting patterned changes in demand like heating needs).

Additionally, we demonstrated that 211 request data reflect both emergent needs (like the incipient COVID-19 pandemic) and capture longer-term secular trends in need (like housing and energy costs). Moreover, conversations with community partners in Maine verified that the trends illustrated in this data were in fact reflected in their experiences.

In addition to timeliness, we examined the 211 request data to ensure they are capable of capturing the needs of the population in sufficient granularity to facilitate a policy response. The 211 call taxonomy includes over 1,900 specific request types, allowing for detailed analysis of community needs. Requests related to a broad category like substance use include 53 more granular taxa. The analysis provides strong evidence that 211 data captures a comprehensive view of service needs across Maine. While, again, there is no single threshold at which we can determine the extent to which these data provide a sufficiently comprehensive assessment of the specifics of community needs, we argue that the sheer volume of need taxa presents compelling evidence of their sufficiency.

Moreover, modeling shows that economic deprivation, rurality, and housing quality are significant predictors of higher 211 request volume, even when controlling for population. This provides evidence that requests reflect underlying community socioeconomic factors. Variation in predictive relationships across geography highlights localized factors shaping community needs.

There are limitations to these analyses. First, the descriptive analyses conducted here are relatively straightforward. There are many ways to examine the geographic, temporal, and qualitative variation in requests for service—we do not claim to have conducted the definitive tests here. However, we argue that the results of our straightforward analyses permit the clear appraisal of the merits of 211 data.

Our inferential analyses have similar limitations. For example, our OLS regression model is underpowered and consequently suffers from multicollinearity. We discourage readers from interpreting the statistical estimates presented as an attempt to precisely quantify the relationships between different community characteristics and call volume, but rather as an attempt to demonstrate the broad relationships between these constructs.

There are similar limitations in the GWR. While Geographically Weighted Regression (GWR) is a powerful tool for capturing spatial heterogeneity, it can be sensitive to multicollinearity among independent variables, potentially leading to unstable parameter estimates (Wheeler & Tiefelsdorf, 2005). They also risk overfitting, particularly in areas with sparse data, which can affect generalizability (Fotheringham, Brunson, & Charlton, 2002). Choosing the correct bandwidth is challenging and crucial for balancing model complexity and accuracy.

Additionally, GWR is computationally intensive, and its assumptions about smooth spatial variations may not always hold true, leading to biased estimates (Páez, Farber, & Wheeler, 2011). Interpretation of local parameters can be complex, and traditional validation techniques may not fully capture spatial variability, potentially leading to overconfidence in model predictions (Brunsdon, Fotheringham, & Charlton, 1998). Again, we encourage readers not to interpret our GWR findings as representing optimal and precise estimates of the relationships between community characteristics and call volume, but rather demonstrating the ease of conducting similar studies and the presence of existing—intuitive—geographic variation in the 211 data.

211's statewide coverage in Maine is an advantage; fragmentation of systems in other states could limit the generalizability of these findings. The anonymous nature of requests also prevents deduplication, which could limit accuracy of analyses that rely on it. Also, differences in state funding, and variation in the level of coordination across state human service departments, shape how systems can respond to surges in demand.

Overall, with cognizance of limitations, 211 service request data provide a valuable lens into community needs at multiple geographic levels in Maine. The previous analyses—we argue—demonstrate the potential for that lens in informing secondary prevention efforts for children and families. Extending best practices on coverage, funding models, and taxonomies could unlock similar potential in 211 systems nationwide.

Conclusion

In the previous section, we demonstrated that 211 Maine data are reasonably comprehensive, timely, and granular. We conclude by discussing their usefulness. As

discussed above, extensive research demonstrates that child welfare system involvement often follows a family coming under new or continuing socioeconomic pressures. Thus, these data's fitness for secondary prevention may allow them to serve as a lead indicator for populations on the cusp of child welfare system involvement. Moreover, their relatively high level of geographic precision may allow for rapid adjustment of resource delivery and more flexible deployment strategies.

Beyond identifying aggregate levels of community stress or resource need, 211 data provide fine-grained information on the specifics of community needs. As highlighted above, this precision enables community providers to calibrate services with a high degree of precision. For example, 211 taxa enables service providers to distinguish between different types of substance use concerns. Moreover, policymakers can use 211 to account for seasonal variation in a community's "mixture" of needs.

The specificity, timeliness, and comprehensiveness of 211 data would, finally, allow policymakers to buttress their existing data toolkits. As demonstrated above, 211 data can be considered alongside data sources like the United States Census, or other administrative data, to develop a variety of insights. These insights could include preliminary evidence of the social, economic, or demographic determinants of community needs. At the very least, cross-referencing 211 service request data with other sources can help policymakers to contextualize insights generated from these other, more traditionally relied upon, sources of information.

Taken together, these attributes provide human service system policymakers with a valuable toolkit to support secondary prevention service delivery. The increasing focus on secondary prevention in the child welfare space necessitates leveraging data sources fit for the task of planning and building secondary prevention service arrays.

Succeeding in this task will require both the coordination of multiple human service systems and information sufficient to guide their decision making. We believe the evidence discussed in this paper makes a compelling case that 211 service request data provides an excellent candidate to do just that.

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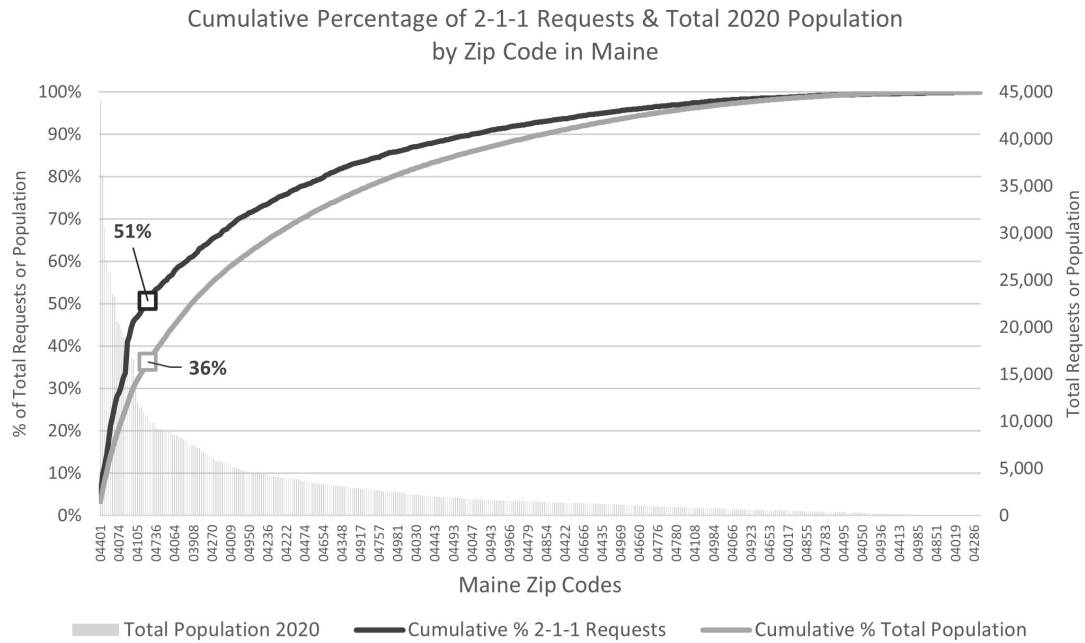


Figure 1. Cumulative Percentage of 211 Requests & Total 2020 Population by ZCTA. ZCTAs are listed in descending population order. Source: 211 Maine data and 2015-2020 American Community Survey.

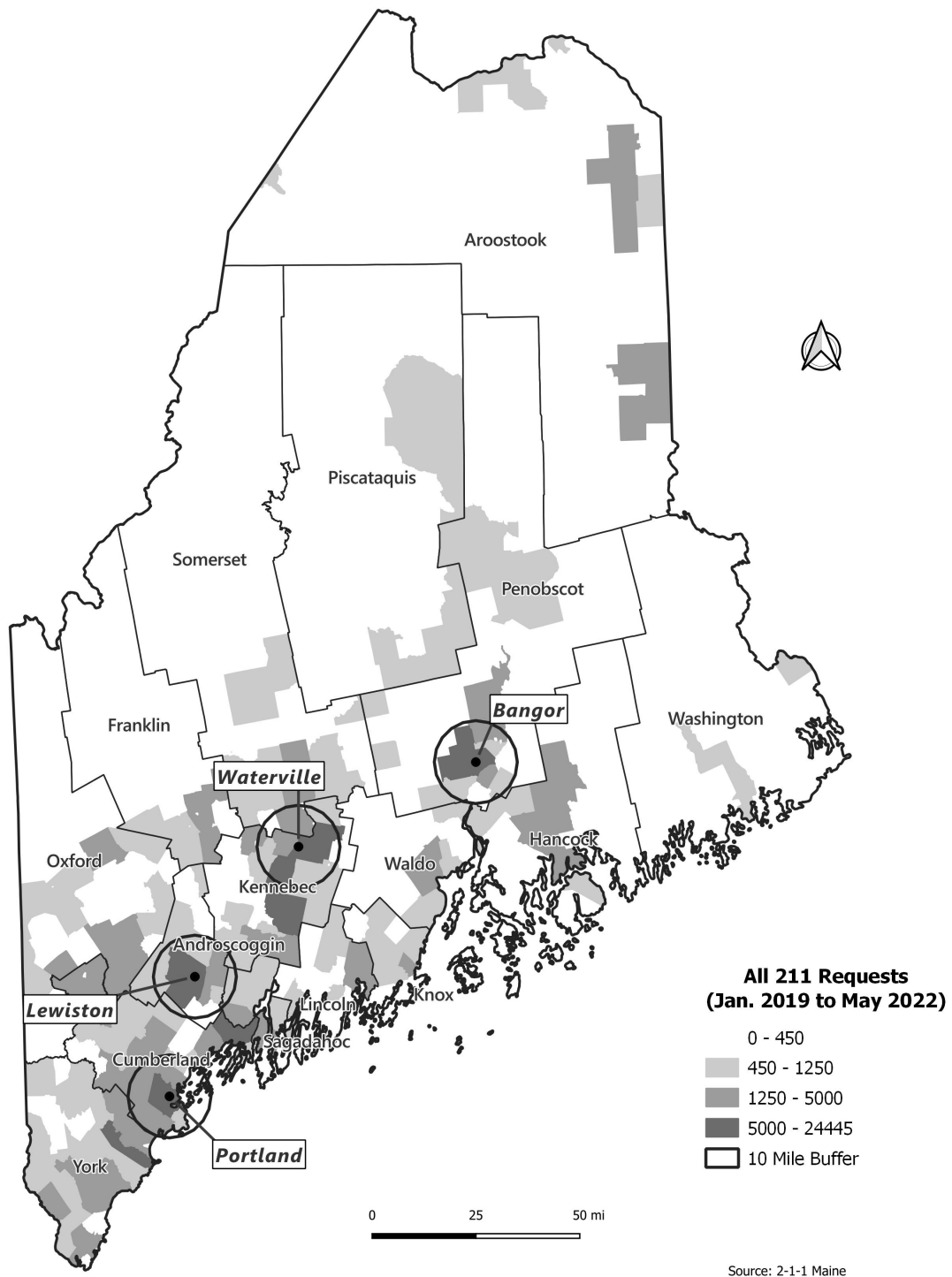


Figure 2. Map of 211 requests by ZIP Code January 2019 to May 2022. Source: 211 Maine Data.

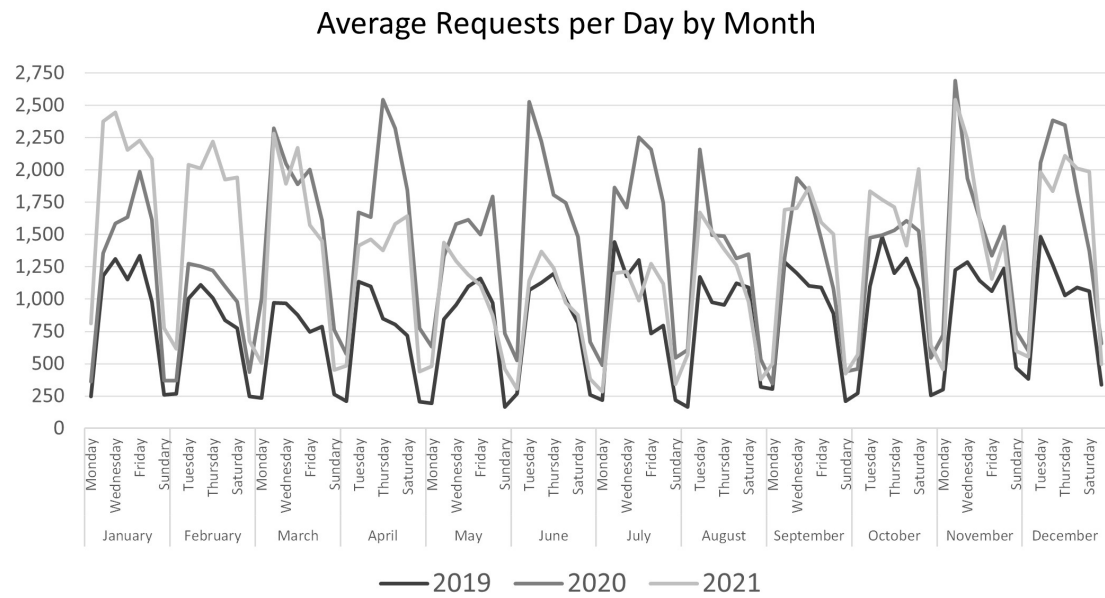


Figure 3. Average Requests Per Day, by Month, 2019-2021. Source: 211 Maine Data.

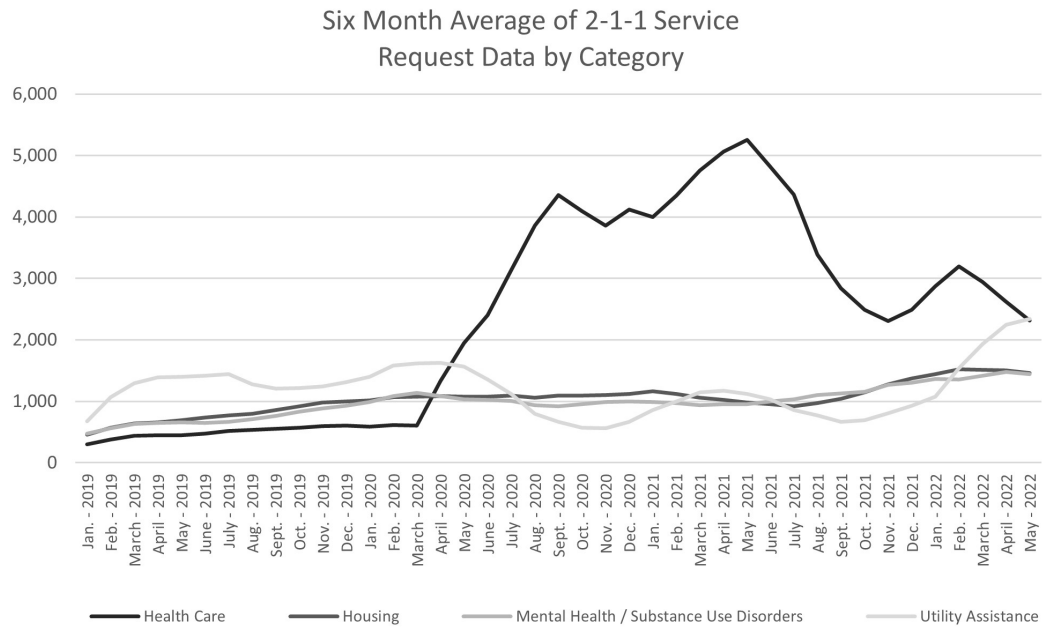


Figure 4. Six Month Rolling Averages of 211 Service Requests by Category, January 2019-May 2022.

Table 1. Coefficient Estimates from OLS Model. Source: 211 Maine data and 2015-2019 5-year ACS estimates.

	Log(Requests)
	calls
	OLS
Population	-0.001 (0.001)
White, NH Population	0.0002 (0.0003)
# with < H.S. Education	0.0002 (0.001)
# with Diploma/GED	-0.0001 (0.0004)
# at same address as last year	0.001* (0.0004)
# under 18	0.00003 (0.001)
# foster children	0.003 (0.002)
# single mothers	-0.0004

	(0.001)
# on SSI	0.0004
	(0.001)
# on Supp. S.I.	0.003*
	(0.001)
# on public assistance	-0.004*
	(0.002)
# on SNAP	-0.0003
	(0.001)
# of veterans	-0.0002
	(0.001)
# of vacant dwellings	0.0001
	(0.0002)
# housing units built <1930	0.001*
	(0.0003)
# of renters	-0.0004
	(0.001)
# housing units w/ oil heat	-0.001*
	(0.0003)
# housing units w/ wood heat	0.003*
	(0.001)

# housing units w/ poor plumbing	-0.001 (0.001)
# uninsured <= age 18	-0.001 (0.001)
# uninsured age 19-34	0.0001 (0.001)
# uninsured age 35-64	0.001 (0.001)
# below 50% FPL	0.001 (0.001)
# at 51-99% FPL	-0.0003 (0.001)
# employed in nat. res. extraction	0.002 (0.001)
# employed in construction	0.002 (0.001)
# employed in manufacturing	0.001 (0.001)
# employed in wholesale	-0.001 (0.002)

# employed in retail	0.0003 (0.001)
# employed in supply chain	-0.001 (0.001)
# employed in service ind.	-0.001 (0.001)
# over 65	-0.001 (0.001)
# aged 14-17 not in HS	-0.004 (0.005)
# aged 3-4 not in Pre-K	-0.001 (0.002)
# aged >= 25 w/ Bachelors+	-0.0001 (0.0004)
# with 1hr+ commute	0.002* (0.001)
# aged 16+ unemployed	-0.001 (0.001)
# 16+ not in labor force	0.0001 (0.001)
# units with 1+ occ./room	-0.002

	(0.002)
# units built on/after 2005	0.0001 (0.0004)
# paying 30% of HHI for housing	0.002* (0.001)
# units valued < 100k	0.001* (0.0004)
# units valued > 500k	0.0003 (0.001)
# HH with 3+ vehicles	0.001 (0.0004)
Constant	4.007* (0.066)

Observations	430
R ²	0.766
Adjusted R ²	0.739
F Statistic	28.623* (df = 44; 385)

Note: * p<0.05

Table 2. Coefficient Estimates from GWR Model.

	Min.	1st Qu.	Median	3rd Qu.	Max	Global
Constant	3.142	3.536	4.020	4.283	4.443	4.036
Population	-0.00	-0.00	-0.002	-0.001	0.001	-0.001
White, NH Population	0.000	0.001	0.001	0.002	0.003	0.000
# with < H.S. Education	-0.00	0.000	0.000	0.001	0.002	0.000
# with Diploma/GED	-0.00	0.000	0.000	0.001	0.002	0.000
# at same address as last year	0.000	0.001	0.001	0.002	0.003	0.001
# under 18	-0.00	0.000	0.001	0.001	0.004	0.000
# foster children	-0.00	-0.00	0.001	0.005	0.009	0.003
# single mothers	-0.00	-0.00	-0.001	0.000	0.007	0.000
# on SSI	-0.00	0.000	0.001	0.003	0.005	0.000
# on Supp. S.I.	-0.00	0.000	0.002	0.004	0.008	0.003
# on public assistance	-0.00	-0.00	-0.001	0.002	0.007	-0.004
# on SNAP	-0.00	-0.00	-0.001	0.000	0.006	0.000
# of veterans	-0.00	0.000	0.001	0.002	0.004	0.000
# of vacant dwellings	-0.00	0.000	0.000	0.001	0.001	0.000
# housing units built <1930	-0.00	0.000	0.000	0.000	0.002	0.001
# of renters	-0.00	-0.00	0.000	0.001	0.003	0.000
# housing units w/ oil heat	-0.00	-0.00	-0.001	-0.001	0.000	-0.001
# housing units w/ wood heat	-0.00	0.001	0.002	0.002	0.004	0.003
# housing units w/ poor	-0.00	-0.00	0.000	0.000	0.002	-0.001
# uninsured <= age 18	-0.00	-0.00	0.000	0.001	0.004	-0.001
# uninsured age 19-34	-0.00	0.000	0.001	0.002	0.006	0.000
# uninsured age 35-64	-0.00	-0.00	0.000	0.000	0.002	0.001
# below 50% FPL	0.000	0.000	0.002	0.003	0.004	0.001
# at 51-99% FPL	-0.00	0.000	0.000	0.001	0.002	0.000
# employed in nat. res. extraction	-0.00	0.001	0.002	0.003	0.008	0.002

# employed in construction	-0.00	0.001	0.001	0.003	0.005	0.002
# employed in manufacturing	-0.00	-0.00	0.001	0.002	0.004	0.001
# employed in wholesale	-0.01	-0.00	-0.004	-0.002	0.003	-0.001
# employed in retail	-0.00	0.000	0.001	0.001	0.002	0.000
# employed in supply chain	-0.01	-0.00	-0.002	0.000	0.006	-0.002
# employed in service ind.	-0.00	-0.00	-0.001	0.001	0.003	-0.001
# over 65	-0.00	-0.00	-0.001	0.000	0.001	-0.001
# aged 14-17 not in HS	-0.03	-0.01	0.002	0.006	0.018	-0.004
# aged 3-4 not in Pre-K	-0.00	-0.00	-0.003	-0.001	0.004	-0.001
# aged >= 25 w/ Bachelors+	-0.00	-0.00	0.000	0.000	0.001	0.000
# with 1hr+ commute	-0.00	-0.00	0.000	0.002	0.003	0.002
# aged 16+ unemployed	-0.00	-0.00	-0.001	0.003	0.011	-0.001
# 16+ not in labor force	-0.00	-0.00	0.000	0.001	0.003	0.000
# units with 1+ occ./room	-0.02	-0.00	-0.004	0.001	0.006	-0.002
# units built on/after 2005	-0.00	0.000	0.000	0.001	0.002	0.000
# paying 30% of HHI for	-0.00	0.001	0.001	0.002	0.005	0.002
# units valued < 100k	-0.00	0.001	0.002	0.002	0.005	0.001
# units valued > 500k	-0.00	-0.00	0.000	0.002	0.014	0.000
# HH with 3+ vehicles	-0.00	0.000	0.001	0.001	0.002	0.001

Quasi-Global R2 = 0.8763

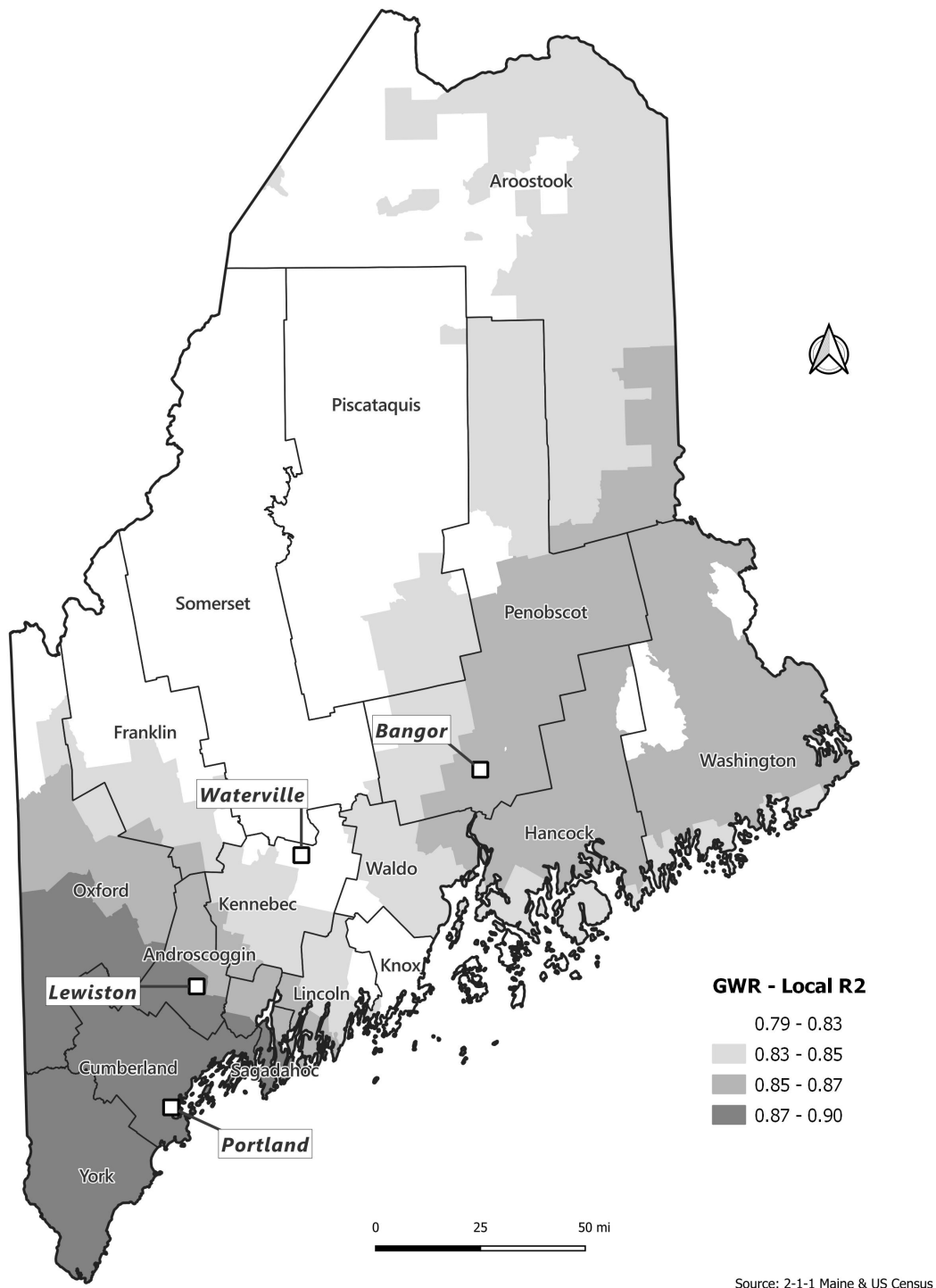


Figure 5. Local variation in R^2 generated from GWR model summarized in Table 2.

Source: 211 Maine and TIGRIS.